

# Product Roadmap (V1 to V4)

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Product: Cubyntra  
Organization: Necookie Labs  
Current Milestone: V1 (Shipped)

## 1. Version Overview Matrix

| Release | Focus Area                                                          | Status             | Target Delivery |
|---------|---------------------------------------------------------------------|--------------------|-----------------|
| V1.0    | Core Client-Side Vision, Validation, Kociemba Solver, 3D Twin       | Completed          | Q3 2026         |
| V2.0    | Web Worker Multithreading, Automatic Homography Warping, Voice Cues | Planned            | Q1 2027         |
| V3.0    | Bluetooth Smart Cube Support, CFOP Training Suite, Reconstructions  | Planned            | Q3 2027         |
| V4.0    | Physical Robotics Rig, ESP32 Kinematics, Web Serial Controller      | Future / Scope Out | 2028            |

## 2. Milestone Details

### 2.1 Version 1.0 (Current Release) — The Solid Foundation

- [x] Client-side Computer Vision using WebRTC camera stream.
- [x] Circular kernel color sampling with RGB -> HSV and CIELAB transforms.
- [x] Delta-E color classification with real-time temporal stability filtering.
- [x] Mathematical invariant validator (cardinality, physical piece integrity, parity).
- [x] Interactive error recovery modal for optical misclassifications.
- [x] Deterministic Herbert Kociemba two-phase solver (< 50ms execution).
- [x] Three.js 3D digital twin with drift-free integer snapping and 3D directional turn arrows.
- [x] Full playback controls (play/pause, step forward/backward, speed multiplier).
- [x] Comprehensive automated test coverage (35/35 passing).
- [x] Complete technical documentation suite and PDF specifications.

### 2.2 Version 2.0 — Computer Vision & Worker Optimization

- [ ] Web Workers Solver: Offload Kociemba IDA\* search to dedicated background workers, guaranteeing 0ms main thread blocking during complex scrambles.
- [ ] Automatic Perspective Homography: Automatic four-corner contour detection and perspective rectification, removing the need for a rigid reticle overlay.
- [ ] Voice-Guided Step Directions: Text-to-speech engine speaking aloud moves (e.g. "Turn the Right face clockwise 90 degrees") for hands-free solving.
- [ ] Ambient Light Auto-Calibration: Dynamic histogram equalization to handle extreme low-light or yellow tungsten environments.

### 2.3 Version 3.0 — Smart Cubes & Advanced Speedcubing

- [ ] Bluetooth Smart Cube Integration: Web Bluetooth API connection to GAN, GoCube, and Moyu smart cubes for real-time hardware telemetry.
- [ ] CFOP & Roux Method Training: Breakdown of solutions into Cross, F2L, OLL, and PLL stages with algorithmic alternatives.
- [ ] URL State Serialization: Encode scramble positions and custom solution paths into compact URL hash fragments for

- zero-backend sharing.
  - [ ] Inspection Timer & Metrics: WCA-compliant 15-second inspection countdown and turns-per-second (TPS) analytics.
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## 2.4 Version 4.0 — Physical Robotics Rig (FUTURE / OUT OF SCOPE FOR V1)

*Physical robotics and microcontroller hardware are strictly categorized as FUTURE / OUT OF SCOPE FOR V1.*

- [ ] ESP32 Firmware: Dedicated micro-ROS or lightweight serial firmware for 6-axis NEMA-17 stepper motor coordination.
- [ ] Web Serial / Web Bluetooth Execution: Direct hardware bridge from Cubyntra web app to the physical robotic rig.
- [ ] Kinematic Acceleration Profiling: S-curve acceleration curves to prevent cube popping during sub-second physical turns.
- [ ] Physical Optical Alignment: Dual-camera stereoscopic rig tracking physical cube faces during mechanical rotation.